Evaluating the Effectiveness of Masked Autoencoders for Computer Vision

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Abstract

In this report, we discuss the approach taken to replicate the results of the paper "Masked Autoencoders Are Scalable Vision Learners" (He et al. [2022]). Our approach involved constructing an asymmetric encoder-decoder architecture with a ViT-B/16 encoder and a simple decoder. After self-supervised pre-training, we focused on supervised training for evaluating the representations with linear probing. We successfully identified and replicated similar trends in the results of ablation tests for varying the masking ratio and decoder block size. We also found a deviating trend in the effect of data augmentation. In addition, Grad-CAM is implemented to observe how the MAE pre-training creates a self-supervisory task that gains a holistic understanding of the image beyond low-level image statistics. These findings along with the various ablation results, are detailed in the report.

1 Introduction

A paper's reproducibility refers to other researchers' ability to obtain the same results as the original authors by following the methods and procedures described in the paper. This report discusses the approach taken to replicate the results of the paper 'Masked Autoencoders Are Scalable Vision Learners'(He et al. [2022]). Self-supervised pertaining with masked autoencoders (MAE) has been very successful in natural language processing (NLP) and led to satisfactory outcomes (i.e. BERT(Devlin et al. [2018])). But there has been a gap in replicating the success with the use of Vision Transformers (ViT) (Dosovitskiy et al. [2020]). The original paper proposed a solution in the vision domain by masking a large portion of an input image and training an asymmetric encoder-decoder architecture to reconstruct the missing pixels.

The overall approach of the original paper can be summarised as building an asymmetric encoder-decoder architecture where the encoder only operates on the visible subset of patches (without mask tokens) and the lightweight decoder reconstructs the original image from the latent representation and mask tokens. The paper performs various ablation tests with the main conclusion that the 75% masking ratio gives the best performance for self-supervisory tasks.

To replicate the results, we follow the same approach as the paper by first constructing an asymmetric encoder-decoder architecture with a ViT-B/16 encoder and a simple decoder with one block depth and 512 node-width. After the initial self-supervised pre-training phase, we continued with supervised training and evaluated the representations with linear probing. End-to-end fine-tuning was not considered in this report due to its computational costs.

We successfully replicated similar trends in the ablation tests by varying the masking ratio and decoder block size. However, we observed a deviating trend in the effect of data augmentation. Similar to the paper's conclusion, we saw a high accuracy using a masking ratio of 75% in linear probing, which in our case was 66.85%.

In addition, we implemented a Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al. [2017]) to see how the MAE pre-training creates a self-supervisory task that gains a holistic understanding of the image beyond low-level image statistics. These findings, along with the various ablation results, are discussed in detail in the upcoming sections. The replicated code can be found at Dadfar and Ganesan [2022].

2 Related Work

MAEs and SSL: Masked auto-encoding has significantly advanced as a promising venture in NLP and computer vision. Inspired by this (Pang et al. [2022]) proposed a neat scheme of masked autoencoders for point cloud self-supervised learning (SSL), addressing the challenges posed by the point cloud's properties, including leakage of location information and uneven information density. (Wu et al. [2022]) proposed a new self-supervised method, which is called Denoising Masked AutoEncoders (DMAE), for learning certified robust classifiers of images. (Feichtenhofer et al. [2022]) presented a simple extension of MAE for spatiotemporal representation learning from videos. Their study suggests that the general framework of masked autoencoding (BERT, MAE, etc.) can be a unified methodology for representation learning with minimal domain knowledge. With all these developments, (Zhang et al. [2022]) conduct a comprehensive survey of masked autoencoders to shed insight on the promising direction of SSL.

3 Methods

The original paper proposes an autoencoder that reconstructs a signal from its partial observation. The approach includes self-supervised pre-training and supervised training for representation evaluation. The implementation is relatively simple and can be broken down into the following steps (see Figure 1). The first step is to mask the input image, where we divide the image into regular, non-overlapping patches and identify a subset of patches (e.g., hidden or masked patches) using a specific sampling strategy (i.e., random sampling) and to remove that subset. This is the step that aids the self-supervised pre-training to learn image representations. The second step is to feed the visible unmasked patches (without mask tokens) into the MAE encoder (e.g. ViT) with their corresponding

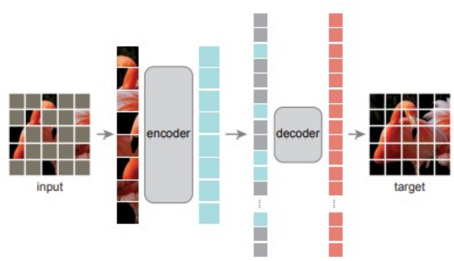


Figure 1: MAE architecture. Image taken from orginal paper (He et al. [2022])

embeddings. In the last step, the full set of tokens consisting of both the encoded visible patches and the mask tokens are assembled together, and after adding the positional embeddings, they are fed into the MAE decoder. Finally, the input image is reconstructed by predicting the pixel values for each masked patch at the end of the decoder. In this approach, The MAE decoder is a series of lightweight transformers that are used only in the pre-training phase to obtain a good image representation. After pre-training, the decoder is discarded, and the encoder is used on a full set of image patches for the supervised recognition tasks.

The original paper uses ViT-L as a baseline comparison in their ablation studies. In our reproduction of the MAE, we have down-scaled the baseline to ViT-B architecture and used a decoder with 1-block-depth and 512-node-width. A need to modify the MAE architecture beyond this scope was not necessary for the reduced computational resources used.

Additionally, we implemented the Grad-CAM to visualize how the MAE understands the image and based on what predictions are made. To generate a Grad-CAM visualization, we first compute the gradient of the chosen layer (here, it is the last layer before the final prediction layer in the decoder) with respect to the predicted class. Then we calculate the weighted sum of the gradient values across all channels of the chosen layer and re-size to the same size as the input image. Finally, a heatmap is generated by overlaying the resized weights on top of the input image.

4 Data

In vision-related tasks, self-supervised learning requires a large quantity of image data. Hence the original paper (He et al. [2022]) uses a large dataset called ImageNet-1K (Deng et al. [2009]) with 1K classes and 1.4M images for MAE pre-training and performance evaluation. But due to the computational resource constraints, we used a downscaled and smaller version of the large ImageNet dataset called Imagenette introduced as of the paper(Howard and Gugger [2020]). More specifically, we used a specific version of the dataset called Imagenette2-320. It is a subset of ImageNet with 10 very different classes and with the same resolution as ImageNet. The dataset has around 1.3K images for each of the classes (In total, it has 13.3K images) which is divided into 70% training and 30% validation sets. Each image is normalized and cropped in both pre-training and later. More details on the data augmentation are discussed in section 5.

5 Experiments and Findings

Table 1: Key default configuration settings for the MAE

configuration	value
masking ratio	75%
decoder block size	1
decoder node size	512
data augmentation	RandomResizeCrop
pre-train epochs	200
linear probing epochs	100

The original paper was exhaustive in its ablation studies and investigates the effects of various parameters, but our investigation will be limited to ablation tests on varying the masking ratio, the decoder depth, and the data augmentation according to our proposal. The key default settings of our MAE are listed in Table 1. We focus on using linear probing rather than fine-tuning due to the time constraints and use the accuracy of the supervised task as the evaluation metric. The results of the experiments using different parameters are displayed in Table 2. The tests are performed with downscaled versions of the MAE and on a smaller dataset, hence the absolute value of the accuracy is lower and different from the paper with a difference of up to 20%. However, the trends are consistent.

Table 2: Results of the ablation experiments varying masking ratio, decoder block size, and data augmentation with 75% masking ratio using linear probing

(a) Masking ratio

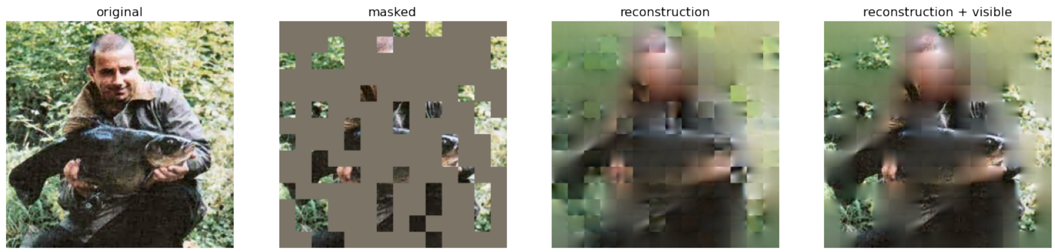
Masking ratio	50%	75%	90%
Top-1 accuracy	64.28%	66.85%	66.29%
Top-5 accuracy	94.75%	95.49%	95.00%

(b) Decoder size and data augmentation with 75% masking ratio

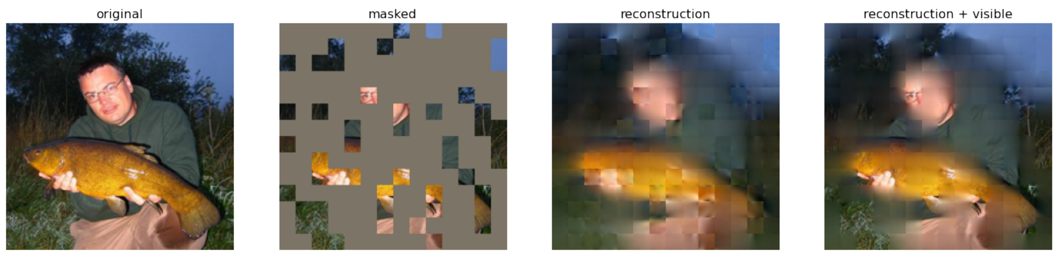
	default	**vary decoder**	**vary augmentation**
Decoder block size	1	8	1
Data augmentation	with	with	without
Top-1 accuracy	66.85%	69.45%	68.33%
Top-5 accuracy	95.49%	96.13%	95.54%

Our baseline model (ViT-B without masking) gives an accuracy of 32.7%. By increasing the masking ratio from the baseline, the accuracy increases steadily until a sweet spot of 75% and decreases afterward. We observed the same trends in the original paper. The difference in the absolute value might be due to using a smaller dataset. This is because the vision transformer requires more data to generalize better. Using our small dataset, the reconstruction quality is not quite good as observed in

Figure 2. The results are expected as the pre-training loss saturates at approximately 0.35 (seen in Figure 3) at the end of 200 epochs. We saw a small improvement in the value of the loss (e.g., 0.33) when we increased the number of epochs to 400.



(a)



(b)

Figure 2: Example outputs from an MAE trained with the default settings

Table 2 shows the effect of increasing the decoder depth. It is observed that the accuracy increases by 2.6% as a result of increasing the decoder blocks from 1 to 8. The same trend is observed in the paper. However, the ablation tests on data augmentation show a deviation from the paper. Our MAE with default settings performs better with no data augmentation (just center-crop and no flipping) and gives an accuracy of 68.33% compared to 66.85% with augmentation (random-size crop and random horizontal flipping). We think this is because data augmentation causes too much disturbance using our smaller dataset. In particular, regularizing the data (using augmentation) did not have significant effects since MAE relies on random masking for the generation of new training samples rather than the data augmentation.

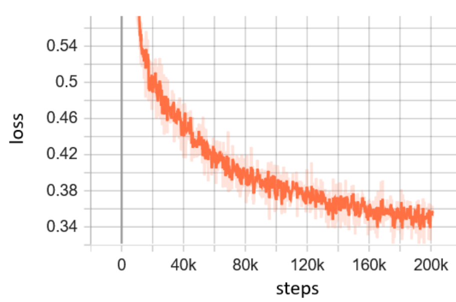
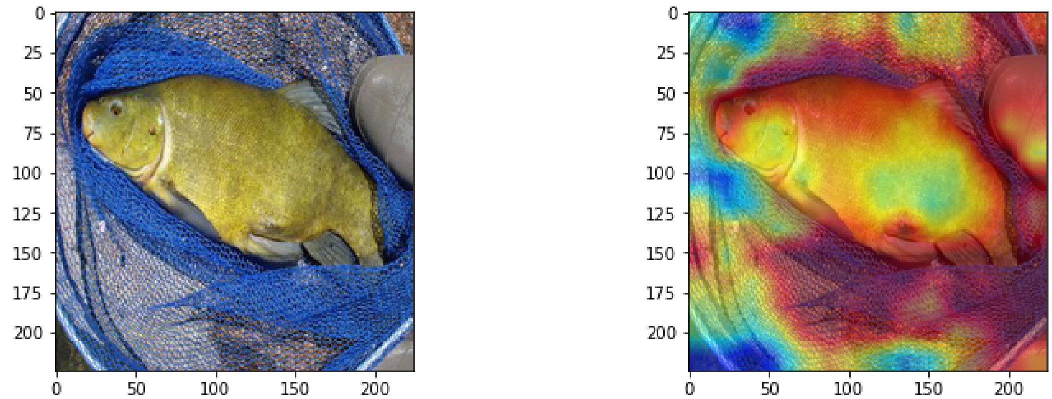


Figure 3: Pretrain loss evolution with one decoder block and masking ratio of 75%.

The visualization of the MAE learning using grad-CAM gives an insight into which part of the image the MAE focuses on. The heatmap on the top of a sample image is seen in Figure 4. The result represents the learning for a sample image from the tench fish class. In this case, the MAE correctly concentrates mostly on the regions of the head, tail, edges, and fins of the fish, but there is also leakage of importance to some surrounding areas. We expect the representation learning and resulting heatmaps of MAE will be better in larger MAE such as ViT-L.



(a)

(b)

Figure 4: Grad-CAM visualization for the MAE learning on an example classified as Tench Fish

6 Challenges

One of the main challenges we faced was to use a smaller dataset for conducting the experiments. We initially decided on Tiny ImageNet(Le and Yang [2015]) but struggled quite a bit to get the MAE to work due to the difference in resolution from ImageNet. Once we made it somehow work, it ended up taking more than 12 hours for just pre-training with the simplest architecture. Then with a bit more research, we used a smaller dataset such as the Imagenette2-320 with the same resolution to replicate the paper. The usage of a smaller dataset cost us in the accuracy metric and seems to influence the study more than any other parameter. Another issue we faced was the lack of clarity in the data augmentation needed for the ablation study. The provided materials in the paper and the appendix were not clear enough. Thus, we were forced to refer to related references. The overall challenge was the computational and time resources that were needed for these experiments which were shared only by two students.

7 Conclusions

This report discussed the approach taken to replicate the results of the MAE paper (He et al. [2022]) under a limited resource constraint environment. The results of the experiments show the accuracy increases steadily with the masking ratio up to 75%. In addition, increasing the decoder depth improves the reconstruction specialization for linear probing. These results are consistent with the paper. However, the ablation tests on data augmentation showed a deviation from the paper, as the MAE with default settings performed better with no data augmentation. Additionally, the Grad-CAM visualization showed that the MAE focuses mostly on the expected regions of the image but with some leakage of importance to the surrounding areas. Thus, the original authors' claim that the MAE learns the rich-hidden representation is entirely feasible.

In the process of replicating the paper, it is certainly necessary to consider the limitations hindering the exact replication. Substantial computational resources and data are needed for replicating such intensive deep-learning experiments. In addition, while the methodology of the original paper was sufficient for coding from scratch, certain nuisances needed to be clearer for replication without referring to the additional references.

Further replication efforts can include adding more parts of the ablation tests, for example changing the sampling strategy, and studying the transfer learning in downstream tasks. Beyond replication, future extensions of this work could include exploring the applications of MAE approach in other domains or in other tasks. Additionally, investigations on how exactly the MAE model can perform the self-supervised pre-training and learns a good representation can be explored.

8 Ethical Consideration, Societal Impact, Alignment with UN SDG targets

The ethical considerations for the original paper are related to the concept of vision learning and their applications in general. These would be ensuring data collected in ethical and responsible manner, ensuring non-violation of privacy concerns in obtaining images for training and ensuring the models using MAE methods are not used for any discriminatory or unethical purposes such as biased facial recognition systems. But other than these generic considerations, the MAE method for self-supervised learning in its methodology or experiments does not pose any issues. There seems to be accurately and transparently reported results from the experiments. It is difficult to determine the potential societal impact of the paper and its possible hinderance or aid to the UN SDG targets for improved deep learning techniques such as MAE for scalable vision learning. This is due to the fact that the impact arises only from the varied applications it can be used for, be it for betterment or worsening of well-being of the society.

9 Self-Assessment

We have successfully reproduced the set of experiments from the original paper that we set out to complete. The results were comparably to that of the original paper given the down-scaling of the MAE capacity and smaller Imagenette dataset. Additionally, we have also included a grad-CAM visualization to further understand the representation learning using MAEs. MAE architecture was built from scratch in PyTorch and can be extended to larger ones with minimal changes. We believe that the work done is in-line with the initial proposal and goes beyond with addition of grad-CAM. So especially given with resources shared by only two team members, this qualifies for the assessment criteria of 'excellent' as in grade A.

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